1.

Module # 1 to 4 and 5-7,

Show whether $f: Z \to Z$ defined by $f(x) = 7x$ is bijective function or not?

Solution:-

Step-I:-

One-to-one function:-

$f(x) = f(y)$

$7x = 7y$

$x = y$

So, f is one-to-one function.

Step-II

Onto function:-

We assume that $x \in Z$ is the pre-image of $B \in Z$. then,

$f(x) = 7x = 8$

$x = \frac{8}{7} \notin Z.$

This implies that f is not bijective.

Note:- If we pick up any integer that is not a multiple of Z 7, then it will not give us any integer in return.

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Module :- B-9.

Q NO:- 01.

Suppose a set A has 6 elements and set B has 8 elements under some binary operation. Can A and B be isomorphic binary structure?

Solution:-

Orders of set A and B are not same. So these two sets don't have one-to-one correspondence. Because for any two sets A and B one-to-one correspondence must exist for them to be isomorphic. Hence, A and B are not isomorphic binary structures.

Q No:- 02

Give the reason whether the binary structure $(C, \cdot)$ and $(R, \cdot)$ under multiplication are isomorphic or not.

Solution:-

The binary structures $(C, \cdot)$ and $(R, \cdot)$ under multiplication are not isomorphic because the equation $x \cdot x = c$ has a solution for all $c \in C$ but $x \cdot x = -1$ has no solution in R.

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4.

Module 10-17.

Q No:- 01.

Is $(Q, \cdot)$ a group under multiplication?

Solution:-

1) Closure property:-

$\forall$ r,s $\in$ Q, then

$r \cdot s \in Q$.

2) Associative property:-

$\forall$ r,s,t $\in$ Q, then

$r \cdot (s \cdot t) = (r \cdot s) \cdot t$

3) Identity property:-

1 is the multiplicative property under multiplication. So,

$\forall$ r $\in$ Q, then $r \cdot 1 \in Q$.

4) Inverse property:-

0 is the multiplicative inverse but $0 \notin Q$.

So, inverse property does not hold in this case.

As inverse property does not hold. Hence $\langle Q, \cdot \rangle$ is not a group.

Question:- 02.

Can a group have more than one identity?

Solution:-

Suppose e and e' are identity elements.

So, $e * x = x * e = x$

$e' * x = x * e' = x$

which holds $\forall$ $x \in G$.

In particular,

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$e = e * e' = e'$

So, $e = e'$

According to the above prove we can say that a group $(G, *)$ has exactly one identity element. Hence, a group has one identity.

Q No:- 03.

Write down the names of any two groups under the binary operation of multiplication.

Solution:-

$(Q - \{0\}, \cdot)$ and $(R - \{0\}, \cdot)$ are two groups under binary operation multiplication.

(SEE TOPIC 11 OF VULMS).

Q No:- 04.

Show that the set of integers $(Z, \cdot)$ is not a group under multiplication.

Solution:-

1) Closure property:-

$\forall\ m, n \in Z$, then $m.n \in Z$.

2) Associative property:-

$\forall\ m, n, p \in Z$, then:

$m.(n.p) = (m.n).p$.

3) Identity property:-

1 is the multiplicative identity element.

So, $\forall\ m \in Z$,

$m.1 = m = 1.m$

4) Multiplicative inverse property:-

Let's say $m = 5 \in Z$, then $m' = \frac{1}{5} \notin Z$

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Since, multiplicative inverse property does not hold under multiplication. So $(Z, \cdot)$ is not a group under multiplication.

Module 18-21.

Q NO:- 01.

Write down the names of any two abelian groups.

Solution:-

$(R, +)$, $GL(n, Q)$ and $(C, +)$.

These are all abelian groups.

(WATCH TOPIC 18 ON VULMS FOR FURTHER DETAILS).

Q NO:- 02.

Is the group $G = \{1, -1, +i, -i\}$ under multiplication abelian or not.

Solution:-

$\cdot$	1	-1	i	$-i$
1	1	-1	i	$-i$
-1	-1	1	$-i$	i
i	i	$-i$	-1	1
$-i$	$-i$	i	1	-1

As commutative property for any of two elements exists i.e.,

$$i \times 1 = 1 \times i.$$

$$-i \times i = +i \times -i$$

So, $G = \{1, -1, i, -i\}$ under multiplication is an abelian group.

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Q No:- 03.

Find the order of each element of the group $\{1, -1, i, -i, j, -j, k, -k\}$.

Solution:-

$1 = 1.1 = 1$

So, $|1| = 1$

$-1 = (-1).(-1) = 1$

So, $|-1| = 2$

$i = i.i = i^2 = -1$

$i = i.i.i = (-1).i = -i$

$i = i.i.i.i = (i^2)(i^2) = (-1)(-1) = 1.$

So, $|i| = 4$

$-i = (-i).(-i) = i^2 = -1$

$-i = (-i).(-i).(-i) = -i^2.i = -(-1).i = i$

$-i = (-i).(-i).(-i).(-i) = 1.$

So, $|-i| = 4.$

$j = j.j = j^2 = -1$

$j = j.j.j = j^2.j = -j$

$j = j.j.j.j = j^2.j^2 = 1$

So, $|j| = 4.$

Similarly, $|-j| = 4$

$|k| = 4$

$|-k| = 4$

Module 22-25.

Q No:- 01.

Is Klien 4 group not a cyclic group?

Solution:-

A group of order 4 is called Klien 4 group.

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in which every element is self-invertible.
i.e. $a = a^{-1}$.

*	e	a	b	c
e	e	a	b	c
a	a	e	c	b
b	b	c	e	a
c	c	b	a	e

Order of a = |a| = 2 , $a*a = a^2 = e$ (Order of element)

Order of b = |b| = 2 $b*b = b^2 = e$

Order of c = |c| = 2. $c*c = c^2 = e$

Order of e = |e| = 1.

So, Klein-4 group is not cyclic because the order of a, b and c is 2 while order of identity |e| is 1. There is no element which can generate the group completely. As there is no generator for Klein-4 group so we can say that it is not cyclic.

~~TOPIC 2~~ (TOPIC 24 ON LMS)

Q NO:- 2.

What is the order of the group $(R, +)$.

Solution:-

Let R be a group under addition, If there is a positive integer n such that $a^n = e$. Then, we call the smallest such positive integer the order of a. So, order of group $(R, +)$ is 'n'.

TOPIC # 26-30.

Q NO:- 01.

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Write down one-step subgroup test;

Solution:-

If S is a subset of the group G, then S is a subgroup of G if and only if S is a non-empty set and whether $a, b \in S$ then $ab^{-1} \in S$.

Q No:- 02 *

Write down the Two-step subgroup test.

Solution:-

A subgroup H of a group G is a subgroup of G if and only if

1) H is closed under binary operation '*' of G.

2) For all $a \in H$ it is true that $a^{-1} \in H$ also

In other words.

Let G be a group and H is a non-empty subset of G. If $a*b$ is in H whenever a and b are in H. and a^{-1} is in H whenever a is in H, then H is a subgroup of G.